Physics of a glider winch launch

Model and numerics of `winch_sim`

In a winch launch a glider is pulled into the air by a long rope. A winch at the far end of the runway reels the rope in. The glider rolls a short distance, lifts off, climbs steeply on the rope and releases near the top, typically at about 40 % of the rope length above the ground.

This document describes the physics behind the simulator in `winch_sim/` and how the equations are solved with diffrax. The model has three coupled parts (Figure 1):

1. the **glider**, a rigid body that moves in the vertical plane. It has three degrees of freedom: forward and upward motion, and pitch;
2. the **rope**, about 1–1.5 km of steel wire or synthetic (Dyneema) rope. It has mass, it stretches, it feels air drag, and at the start it lies on the runway;
3. the **winch**, a drum that reels the rope in. It is either an ideal tension-controlled winch or a winch with a power-limited engine.

At the start the glider stands still on the runway and the whole rope lies straight on the grass between the glider and the winch.

Two simple control laws close the loop: the **pilot** moves the elevator, and the **winch driver** sets the throttle.

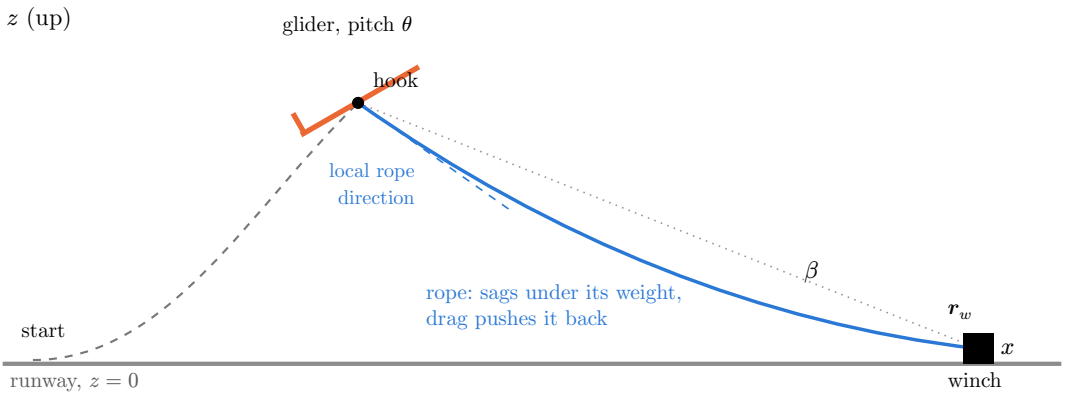


Figure 1: Geometry of a winch launch (not to scale). The glider starts at the left and the winch pulls from the right. The rope sags below the straight line of sight, so at the hook it pulls more steeply downwards than the line to the winch. β is the elevation of the glider seen from the winch.

1 Key physics and assumptions

This section describes the model in words: what happens during a launch, which effects are included, what is simplified or left out, and which inputs matter most for the release height. The equations are in the sections that follow.

1.1 A winch launch in brief

At the start the glider rests tail-down on its main wheel and tail skid. The rope lies straight along the runway between the winch and the hook, just taut: the driver has taken up the slack. After one second the winch driver opens the throttle over six seconds. The rope stretches, and the glider accelerates. The rope that is still on the grass is dragged along with it, which costs some of the pull.

The pilot holds a little back stick during the ground roll. After lift-off the pilot rotates gradually into a steep climb, reaching a pitch attitude of 35–40° by about 30 m height. In the climb the wing carries

the glider's weight *and* the downward pull of the rope, so the wing load factor rises to about 3. The pilot controls the speed with the pitch attitude: nose up if too fast, nose down if too slow.

As the glider gets high above the winch, its elevation seen from the winch, β (Figure 1), increases. From $\beta = 55°$ the pilot starts to lower the nose. Between 65° and 70° the winch driver closes the throttle completely. The drum stops pulling and the rope goes slack. Its weight and the drag of the parachute at its end pull the hook down and back. The glider flies on over the rope end until the cable pulls from behind, at more than 110° below the fuselage axis, and the hook's back-release opens. This is the normal end of a launch in the model. A launch can also end when the weak link breaks, when the rope is almost fully reeled in, or (optionally) when the pilot pulls the release at a chosen cable angle. If the glider has not left the ground after 120 s, the launch is counted as failed (Section 7).

1.2 What the model includes

The glider is a rigid body that moves in the vertical plane: forward, up and in pitch (Section 3). Pitch matters because the rope does not pull through the centre of gravity (CG). The hook sits about 0.3 m ahead of and 0.45 m below the CG (Section 3.2). While the rope is shallow, its pull below the CG pitches the nose up; this is the pitch-up tendency just after lift-off. Once the rope is steeper than about 50–60° below the fuselage axis, the moment changes sign and pitches the nose down, so the pilot needs more and more back stick in the climb. The glider touches the ground at three points (nose skid, main wheel, tail) with springs, dampers and friction (Section 3.3). The wheel rolls with little friction; the skids slide.

Aerodynamics (Section 3.1). Lift grows linearly with the angle of attack, the elevator deflection and the pitch rate, up to a smooth ceiling at the maximum lift coefficient: the wing stalls, but gently. Drag is a parabolic polar with extra drag beyond the stall. The pitching moment gives static stability and pitch damping. All coefficients are estimated from handbook data (span, wing area, best glide ratio, maximum lift coefficient) and generic rules, not measured.

The air. The density falls with height following the International Standard Atmosphere, with an optional airfield elevation and temperature offset. The aerodynamic forces depend on the dynamic pressure, i.e. on the indicated airspeed (IAS); the motion depends on the true airspeed (TAS). The pilot flies IAS, as a real pilot does. An optional headwind grows with height (a power law) and acts on the glider, the rope and the parachute.

The rope is a chain of point masses joined by straight, elastic segments, about 100 m long by default (Section 4, Section 4.1). Four effects are included:

- *Weight.* The rope sags below the straight line to the winch, so the local rope angle at the hook is steeper than the line of sight. The hanging rope pulls the hook down, and the tension at the hook is larger than at the winch by roughly the weight of the rope between the two heights (Section 4.6).
- *Stretch.* The rope acts like a stiff bungee (Section 4.2). It shapes the tension oscillations during the ground run, but hardly changes the energy that reaches the glider.
- *Aerodynamic drag* across and along each segment (Section 4.3). It grows towards the top of the launch, when the upper rope swings round the winch at high speed, broadside to the air.
- *Ground friction* of the rope lying on the grass, which matters for heavy and long ropes (Section 4.8).

The rope end at the glider (parachute, strop and weak link, 8 kg with 0.1 m² of drag area) hangs at the hook. The rope can be steel, Dyneema or any rope given by its datasheet (Section 4.7).

The winch is a drum with an effective mass that reels the rope in against the rope tension and some drum friction (Section 5). Two pull curves are available. The *tension winch* pulls with a fixed force, by default 1.1 times the glider's weight. The *engine winch* has a power limit (200 kW), so its pull drops as the rope speeds up and rises when the glider slows the rope down.

Pilot and winch driver are simple, smooth control laws (Section 5, Section 6). The pilot aims for a pitch attitude that depends on height, IAS and β, and moves the elevator through a PID controller

with a deflection limit and a 0.25 s lag. The winch driver follows a fixed throttle schedule: ramp up at the start, close as β passes 65–70°. The driver does not watch the glider's speed or the rope tension. After release the drum is braked.

1.3 Main assumptions and simplifications

Table 1 lists the main simplifications. Most are standard for a longitudinal flight model. The pilot and winch-driver laws are the weakest part: real people react later, less smoothly and less predictably.

Table 1: Main assumptions of the model.

topic	**assumption in the model**
motion	2-D, longitudinal only: no roll, yaw, sideslip or crosswind
glider	rigid body; no wing bending, no flaps, no ground effect
aerodynamics	quasi-steady coefficients, linear up to a smooth stall; generic estimates from handbook data, not manufacturer data
air	ISA density with height; IAS = equivalent airspeed (no instrument or position error, no compressibility); steady wind, no gusts or thermals
rope	point masses joined by massless, straight, linearly elastic segments; rope reeled in by shrinking every segment equally, so mass leaves all nodes instead of only at the drum
rope end	parachute, strop and weak link lumped at the hook; closed parachute with a constant drag area; not modelled after release
contact	smooth spring–damper ground contact and smoothed Coulomb friction, for the glider and the rope
winch	one effective drum mass; a smooth pull curve (constant force or power-limited)
pilot	attitude control with a smooth schedule and a PID elevator law; no reaction delay beyond the 0.25 s stick lag
winch driver	open-loop throttle schedule in time and glider elevation β; no reaction to speed or tension
end of launch	back-release at a fixed cable angle of 110° below the fuselage axis; weak link breaks at a fixed tension

Every switch-like effect (the rope going slack, a wheel leaving the ground, friction, the stall, the phases of the pilot's schedule) is rounded off smoothly. This keeps the equations one continuous system without mode changes, which makes batching and exact gradients possible (Section 8). Near the switch points, the smoothing is a small approximation.

Not modelled: ground effect, flaps, lateral motion and wing drop, cable breaks other than the weak link, the parachute opening after release, gusts and turbulence, instrument errors, human reaction delays and mistakes, and the detailed mechanics of the winch drum and engine. The glider data are approximations. See Section 11 for more.

1.4 What matters most for the release height

The simulator computes the gradient of the release height with respect to every parameter (Section 10). Table 2 shows the largest ones for the default launch: ASK 21, 1200 m Dyneema rope, tension winch, no wind. The release height is 489 m.

Table 2: Largest sensitivities of the release height; ASK 21, 1200 m Dyneema, tension winch ($h = 489$ m). Elasticity: % change in height per % change in the parameter. Local values; each parameter is changed on its own.

parameter (default)	**Δh for +10 %**	**elasticity**
rope length (1200 m)	+47 m	0.97
winch pull (1.1 × weight)	+26 m	0.52
glider mass (470 kg)	−22 m	−0.46
pilot: β where nose-lowering ends (70°)	+8 m	0.15
pilot: climb attitude (40°)	+6 m	0.12
pilot: target speed (105 km/h IAS)	−6 m	−0.12
pilot: β where nose-lowering starts (55°)	+4 m	0.09
back-release angle (110°)	+4 m	0.09
rope diameter (5 mm)	−3 m	−0.07
rope normal drag coefficient (1.2)	−3 m	−0.07
wing area (17.95 m²)	+3 m	0.06

The picture is simple. The energy the winch puts into the rope is roughly its pull times the length of rope reeled in, so rope length and winch pull come first. A heavier glider needs more of that energy per metre of height, so mass comes next, with the opposite sign. Then come the pilot's and the driver's choices near the top: when to lower the nose, how steep to climb, how fast to fly. These change the height by a few metres each, and their signs can differ between set-ups (with a steel rope, for example, the back-release angle has the opposite effect). The rope's drag (diameter × drag coefficient) is the most important rope property. The weight of a Dyneema rope costs well under a metre; for steel, five times heavier, it is comparable to the drag (about −3 m for +10 %). Rope stiffness has practically no effect. The glider's aerodynamics (polar, maximum lift) change the height by only about 1–2 m for +10 %.

Some inputs have no simple percentage because their default is zero. A headwind adds about 22 m of height per m/s at 10 m. With the tension winch, a high or hot airfield costs very little height (about 2 m per 1000 m of elevation), because the pull is the same and the pilot flies the same IAS; the glider only needs more true airspeed and a longer ground roll. These are local derivatives around the default launch. For large changes, run the simulation directly.

2 Frames, state and conventions

The whole launch happens in one vertical plane, along the runway. We assume that nothing moves sideways: there is no crosswind, no roll and no yaw. The runway is flat and level.

Earth frame x is horizontal and points from the glider's start towards the winch; z points up. The runway is $z = 0$. The rope leaves the winch at $\boldsymbol{r}_w = (x_w, z_w)$, where $z_w = 1$ m (`Winch.height`) and x_w is the hook's start position plus the laid-out rope length (`Launch.rope_length`).

Body frame x_b points forward along the fuselage, z_b points up towards the canopy. In earth coordinates the body axes are $\hat{\boldsymbol{x}}_b = (\cos\theta, \sin\theta)$ and $\hat{\boldsymbol{z}}_b = (-\sin\theta, \cos\theta)$.

Signs The pitch attitude θ, the pitch rate q and the pitching moment M are positive nose-up. In the plane, a force $\boldsymbol{F}$ that acts at offset $\boldsymbol{r}$ from the centre of gravity (CG) gives the moment $M = \boldsymbol{r} \times \boldsymbol{F} = r_x F_z - r_z F_x$.

Points on the glider A point with body coordinates $\boldsymbol{b}$ (hook, wheel, skid) sits at $\boldsymbol{r}_{\text{cg}} + \boldsymbol{r}$ with $\boldsymbol{r} = R(\theta)\boldsymbol{b}$, the body vector rotated by θ. It moves with velocity $\boldsymbol{v} + q\boldsymbol{r}^{\perp}$, where $\boldsymbol{r}^{\perp} = (-r_z, r_x)$ is $\boldsymbol{r}$ turned by +90°.

Smooth functions. Many things in a launch switch on or off: a rope goes slack, a wheel leaves the ground, the wing stalls. Instead of sharp switches the model uses smooth versions, so the equations have no kinks (Section 8). They are:

- $\mathrm{softplus}_s(x) = s\ln\bigl(1 + e^{x/s}\bigr)$, a rounded $\max(x, 0)$; the width s sets how sharp the corner is;
- $\mathrm{softclip}(x, a, b) = a + \mathrm{softplus}_s(x - a) - \mathrm{softplus}_s(x - b)$, a rounded clip of x to $[a, b]$;
- the logistic function $\sigma(x) = 1/(1 + e^{-x})$, a smooth step from 0 to 1;
- the smoothstep $S(x) = 3x^2 - 2x^3$ for $0 \le x \le 1$ (0 below, 1 above), used for gradual changes such as the throttle ramp.

Table 3: State vector y (SI units).

symbol	meaning
$\boldsymbol{r} = (x, z)$, $\boldsymbol{v}$	glider CG position and velocity (earth frame)
θ, q	pitch attitude and pitch rate
$\boldsymbol{p}_i$, $\boldsymbol{u}_i$, $i = 1, ..., N-1$	positions and velocities of the free rope nodes
L_0	unstretched length of the rope still paid out
v_r	reel-in speed at the drum
δ_e	elevator deflection
e_I	time integral of the pilot's attitude error

With N rope segments the state has $6 + 4(N-1) + 4$ entries. For the default 1200 m rope ($N = 12$) that is 54. We use earth-frame velocities for the glider instead of the more usual body-frame velocities. For motion in a plane the two are equivalent, and the earth frame makes the coupling to the rope nodes simpler.

Initial state. The glider stands still on its main wheel and tail skid, tail down. The rope lies straight on the runway between the hook and the winch. Its unstretched length is the distance between the two ends times $1 + s$, with the relative slack $s =$ `Launch.slack`. The default $s = 0$ starts the launch after the driver has taken up the slack: the rope is just taut and carries no tension. With slack the drum spins up freely until the rope comes taut, and then snatches the glider.

3 Glider rigid body

The glider is pulled by the rope, lifted and slowed by the air, pulled down by gravity and, at first, held up by the ground. These forces set how fast it accelerates and climbs. Where they act relative to the CG sets how it pitches.

Assumptions. The glider is a rigid body of constant mass; the wings and fuselage do not bend. Only the longitudinal motion is modelled (three degrees of freedom: x, z, θ). The rope end assembly at the hook (parachute, strop, weak link) moves with the glider: its mass is added to the glider's mass for translation, but not to the moment of inertia. Its weight and the parachute drag act at the hook, so they do pitch the glider.

Newton's laws for translation and rotation read

$$m_{\mathrm{tot}}\dot{\boldsymbol{v}} = \boldsymbol{F}_{\mathrm{aero}} + \sum_k \boldsymbol{F}_{\mathrm{gnd},k} + \boldsymbol{F}_{\mathrm{hook}} - mg\hat{\boldsymbol{z}} \tag{1}$$

$$I_{yy}\dot{q} = M_{\mathrm{aero}} + \sum_k \boldsymbol{r}_k \times \boldsymbol{F}_{\mathrm{gnd},k} + \boldsymbol{r}_{\mathrm{hook}} \times \boldsymbol{F}_{\mathrm{hook}}, \quad \dot{\theta} = q, \quad \dot{\boldsymbol{r}} = \boldsymbol{v} \tag{2}$$

with

- m: launch mass of the glider (with pilots); $g = 9.81\,\mathrm{m/s^2}$ (`Env.g`; the ISA formulas below use the standard $g_0 = 9.80665\,\mathrm{m/s^2}$); $\hat{\boldsymbol{z}} = (0, 1)$;

- $m_{\text{tot}} = m + m_{\text{end}}$: the glider plus the rope mass carried at the hook, $m_{\text{end}} = m_{\text{assy}} + \frac{1}{2}m_n$ (end assembly of 8 kg plus half a rope segment, Section 4.5). The weight of m_{end} is part of $\boldsymbol{F}_{\text{hook}}$; after release $m_{\text{end}} = 0$;
- I_{yy}: pitch moment of inertia about the CG;
- $\boldsymbol{F}_{\text{aero}}$, M_{aero}: aerodynamic force and pitching moment (Section 3.1);
- $\boldsymbol{F}_{\text{gnd},k}$: force of the ground on contact point k at offset $\boldsymbol{r}_k$ (Section 3.3);
- $\boldsymbol{F}_{\text{hook}}$: force of the rope on the hook at offset $\boldsymbol{r}_{\text{hook}}$ (Section 3.2, Section 4.5).

3.1 Aerodynamics

The wing needs airspeed to lift. The glider's speed through the air, and the angle at which the air meets the wing, set the lift, the drag and the pitching moment.

Wind. A headwind shortens the ground run. It grows with height as a power law,

$$\boldsymbol{w}(z) = (-W_{10}(z_c/10\,\text{m})^p, 0), \quad z_c = \text{softplus}_{0.5\,\text{m}}(z) + 0.1\,\text{m}, \tag{3}$$

where W_{10} is the headwind at 10 m (`Env.wind_ref`, default 0) and $p = 1/7$ (`Env.wind_exp`). A headwind blows towards $-x$. The height z_c is kept just above zero so the power law stays finite on the ground. The glider feels the wind at its CG height.

Air data. The air-relative velocity is $\boldsymbol{v}_a = \boldsymbol{v} - \boldsymbol{w}(z)$. The true airspeed (TAS) is $V = |\boldsymbol{v}_a|$. With $u = \boldsymbol{v}_a \cdot \hat{\boldsymbol{x}}_b$ and $w_b = \boldsymbol{v}_a \cdot \hat{\boldsymbol{z}}_b$ the airflow components along the body axes,

$$\alpha = \text{atan2}(-w_b, u), \quad \bar{q} = \frac{1}{2}\rho(z)V^2, \quad \hat{q} = \frac{q\bar{c}}{2\sqrt{V^2 + 1\,\text{m}^2/\text{s}^2}}, \tag{4}$$

where α is the angle of attack of the fuselage axis, $\bar{q}$ the dynamic pressure, $\hat{q}$ the non-dimensional pitch rate and $\bar{c}$ the mean chord. The $1\,\text{m}^2/\text{s}^2$ under the root keeps $\hat{q}$ finite at rest.

Air density. Thinner air gives less lift at the same true airspeed. The density follows the International Standard Atmosphere (ISA, `atmosphere.py`) at the pressure height $h = h_{\text{field}} + z$, with a temperature offset ΔT for hot or cold days (`Env.field_elevation`, `Env.isa_dT`; CLI `--field-elevation`, `--isa-dt`):

$$T = T_0 - \lambda h + \Delta T, \quad p = p_0\left(\frac{T_0 - \lambda h}{T_0}\right)^{g_0/(R\lambda)}, \quad \rho = \frac{p}{RT} \tag{5}$$

with $T_0 = 288.15$ K, $p_0 = 101325$ Pa, lapse rate $\lambda = 6.5$ K/km, $g_0 = 9.80665\,\text{m/s}^2$ and gas constant $R = 287.05\,\text{J/(kg K)}$. Above 11 km the temperature is constant. The offset ΔT changes the temperature and so the density, but not the pressure. The same $\rho(z)$ is used for the glider, for every rope segment (at its mid-height) and for the parachute. At 3000 m the density is 26 % lower than at sea level.

Indicated and true airspeed. Pilots, instruments and placards use the indicated airspeed (IAS). We neglect instrument error and compressibility (fine below about 300 km/h). Then IAS equals the equivalent airspeed, the speed that gives the same dynamic pressure at sea-level density:

$$V_{\text{IAS}} = V\sqrt{\rho(z)/\rho_0}, \quad \rho_0 = 1.225\,\text{kg/m}^3. \tag{6}$$

Stall and the aerodynamic loads depend on $\bar{q}$, and so on IAS. The pilot's target speed and the placarded maximum winch-launch speed V_W are therefore IAS values, while the motion itself uses TAS. The summaries report maximum and minimum IAS (to compare with V_W and the stall speed) and maximum TAS. On a hot, high airfield the glider needs more TAS for the same IAS, and so a longer ground run.

Aerodynamic coefficients. The lift coefficient grows linearly with the angle of attack until it levels off at stall. Drag has a constant part and a part that grows with lift (induced drag); past the stall it rises steeply. The pitching moment keeps the glider stable and is changed with the elevator:

$$
\begin{aligned}
C_L &= \operatorname{softclip}\big(C_{L0} + C_{L\alpha}\alpha + C_{L\delta}\delta_e + C_{Lq}\hat{q},\ C_{L,\min},\ C_{L,\max}\big) \\
C_D &= C_{D0} + kC_L^2 + 1.3\sin^2(\min(\Delta\alpha, 1.5)), \quad k = \frac{1}{\pi e\ \mathrm{AR}} \\
C_m &= C_{m0} + C_{m\alpha}\alpha_m + C_{mq}\hat{q} + C_{m\delta}\delta_e
\end{aligned}
\tag{7}
$$

- C_{L0}, $C_{L\alpha}$: lift at zero angle of attack and lift slope. The angle α is measured from the fuselage axis, so C_{L0} contains the wing's incidence.
- $C_{L\delta}$, C_{Lq}: lift from elevator deflection and pitch rate.
- $C_{L,\min}$, $C_{L,\max}$: negative and positive stall limits. The soft clip (width 0.05) gives a flat top in the lift curve at stall while staying differentiable.
- C_{D0}: zero-lift drag. kC_L^2 is induced drag, with Oswald factor $e = 0.85$ and aspect ratio $\mathrm{AR} = b^2/S$.
- $\Delta\alpha$: how far α is past the stall, in either direction, $\Delta\alpha = \operatorname{softplus}_{0.02}(\alpha - \alpha_s) + \operatorname{softplus}_{0.02}(\alpha_n - \alpha)$ with $\alpha_s = (C_{L,\max} - C_{L0})/C_{L\alpha}$ and $\alpha_n = (C_{L,\min} - C_{L0})/C_{L\alpha}$. The post-stall term $1.3\sin^2$ is flat-plate drag; it is capped at $\Delta\alpha = 1.5$ rad.
- C_{m0}, $C_{m\alpha}$, C_{mq}, $C_{m\delta}$: pitching-moment coefficients about the CG. The moment uses $\alpha_m = \operatorname{softclip}(\alpha, -0.4, 0.4)$ (about $\pm 23°$), so the moment curve goes flat far beyond the stall.
- δ_e: elevator deflection. Negative δ_e (trailing edge up) pitches the nose up, since $C_{m\delta} < 0$.

Lift acts perpendicular to the airflow, drag along it:

$$
\boldsymbol{F}_{\text{aero}} = \bar{q}S(C_L\hat{\boldsymbol{e}}_v^{\perp} - C_D\hat{\boldsymbol{e}}_v), \quad \hat{\boldsymbol{e}}_v = \boldsymbol{v}_a/V, \quad M_{\text{aero}} = \bar{q}S\bar{c}C_m,
\tag{8}
$$

where S is the wing area and $\hat{\boldsymbol{e}}_v^{\perp}$ is $\hat{\boldsymbol{e}}_v$ turned by $+90°$. There is no ground effect.

Coefficients from handbook data. Flight manuals give mass, span, wing area, best glide ratio and the placarded winch speed, but not the stability derivatives. `gliders.make_glider` fills these in with standard estimates:

- mean chord $\bar{c} = S/b$;
- C_{D0} from the best glide ratio: $(L/D)_{\max} = 1/\left(2\sqrt{C_{D0}k}\right)$;
- $C_{L\alpha}$ from the Helmbold formula $2\pi\ \mathrm{AR}\ /\left(2 + \sqrt{\mathrm{AR}^2 + 4}\right)$, plus 8 % for the tailplane;
- $C_{L0} = 0.35$ and $C_{L,\min} = -0.6$ for all gliders;
- $C_{m\alpha} = -C_{L\alpha} \cdot 0.15$ (static margin 15 % of $\bar{c}$);
- C_{mq}, $C_{m\delta}$, $C_{L\delta}$, C_{Lq} from a tailplane with lift slope 4/rad, tail volume 0.5, tail arm $l_t = 4.5$ m and elevator effectiveness 0.5: $C_{mq} = -3.6\,l_t/\bar{c}$, $C_{m\delta} = -1.0$, $C_{L\delta} = \bar{c}/l_t$, $C_{Lq} = 3.6$;
- C_{m0} so the glider trims at its best-glide lift coefficient $\sqrt{C_{D0}/k}$ with neutral elevator;
- $I_{yy} = mk_y^2$ with a radius of gyration k_y of 1.1–1.4 m.

Table 4 lists the catalogue. The values are approximate and meant to show the differences between classes of gliders, not to certify anything.

Table 4: Glider catalogue (`gliders.CATALOGUE`, approximate data): launch mass, span, wing area, best glide ratio, maximum lift coefficient, placarded winch speed (IAS), weak-link rating and radius of gyration.

glider	m	b	S	$(L/D)_{\max}$	$C_{L,\max}$	V_W	weak link	k_y
Ka 8	290 kg	15 m	14.15 m²	27	1.30	100 km/h	5.0 kN	1.10 m
ASK 13	460 kg	16 m	17.50 m²	27	1.40	120 km/h	10.0 kN	1.30 m
LS4	360 kg	15 m	10.50 m²	40	1.40	130 km/h	6.0 kN	1.15 m
Discus CS	330 kg	15 m	10.58 m²	42	1.40	150 km/h	6.0 kN	1.15 m
ASK 21	470 kg	17 m	17.95 m²	34	1.35	150 km/h	10.0 kN	1.35 m
ASG 29	420 kg	18 m	10.50 m²	52	1.45	150 km/h	8.5 kN	1.20 m
DG-1000	620 kg	20 m	17.53 m²	46	1.40	150 km/h	10.0 kN	1.40 m

3.2 Tow hook

The rope is attached to the winch hook (often called the CG hook). It sits a little ahead of and below the CG, at body coordinates $\boldsymbol{b}_{\text{hook}} = (b_x, b_z) = (0.25, -0.45)$ m (0.35 m forward for the two-seaters ASK 13, ASK 21 and DG-1000). Because the hook is not at the CG, the rope pull also pitches the glider. With rope tension T pulling at angle φ below the fuselage axis, the moment of the rope pull about the CG is

$$M_{\text{hook}} = T(|b_z| \cos\varphi - b_x \sin\varphi). \tag{9}$$

While the rope is shallow (φ small) this moment is nose-up. This is the well-known pitch-up tendency right after lift-off. Once $\tan\varphi > |b_z|/b_x$ (above about 52° to 61° here), in the steep part of the climb, the moment turns nose-down. So the pilot has to hold more and more back stick as the launch goes on.

The hook also has a **back-release**: it opens by itself when the rope pulls from far behind, more than 110° below the fuselage axis. This ends the launch (Section 7).

3.3 Ground contact

Before lift-off the glider rests on the ground. It touches it at three points: the nose skid, the main wheel and the tail wheel or skid. The ground carries the weight and the wheels and skids add friction, which the rope has to overcome during the ground run.

Assumptions. Each contact point is a spring with a damper that can only push, not pull. Friction follows Coulomb's law (friction force proportional to the normal force), rounded near zero speed:

$$N_k = \sigma\left(\frac{\delta_k}{2\,\text{mm}}\right) \text{softplus}_{1\,\text{N}}(k_k\delta_k - c_k\dot{z}_k), \quad F_{x,k} = -\mu_k N_k \tanh\left(\frac{\dot{x}_k}{0.1\,\text{m/s}}\right) \tag{10}$$

- $\delta_k = -z_k$: how far point k is below the runway;
- $\dot{x}_k$, $\dot{z}_k$: velocity of the point;
- k_k, c_k: spring stiffness and damping. Skids: $4 \cdot 10^4$ N/m and 2000 N s/m; main wheel: $25mg$ per metre and $20m$ N s/m;
- μ_k: friction coefficient, 0.4 (nose skid), 0.05 (main wheel, rolling), 0.3 (tail);
- the logistic factor σ fades the force out within a few millimetres above the ground.

The contact points sit at body coordinates $(1.8, -0.55)$ m (nose skid), $(0.2, -0.8)$ m (main wheel) and $(-5.0, -0.4)$ m (tail). At rest the glider sits tail down on the main wheel and tail, at about 4.4° nose-up. This one smooth force law covers standing, the ground roll, lift-off and a touchdown, should it happen. There is no switching between modes, which keeps the equations smooth and makes `jit`, `vmap` and `grad` straightforward.

4 The rope: lumped masses with elasticity, weight and drag

The rope is a large part of the launch. A steel rope weighs about 0.08 kg/m, so 1200 m of it weigh about 100 kg, more than the pilot. At 30 m/s its air drag is hundreds of newtons. It stretches by about 0.5–1 % at a working load of 5–6 kN, so it acts like a long bungee. All three effects change the tension that reaches the hook, the angle at which the rope pulls on the hook, and the oscillations in the ground run.

4.1 Discretisation

Assumptions. We model the rope as a chain of point masses (nodes) joined by straight, massless segments that act as springs with dampers. All the rope's mass sits in the nodes. The rope does not resist bending. Air drag is worked out per segment and shared between its two end nodes.

The paid-out rope is split into N segments of equal rest length $\ell_0 = L_0/N$. Node 0 is fixed at the winch. Nodes $1, ..., N-1$ are free point masses $m_n = \mu\ell_0$, where μ is the rope's mass per metre. Node N is the hook, a point on the glider.

The winch reels in at speed v_r, so $\dot{L}_0 = -v_r$. In the model, *all* rest lengths and node masses shrink together. This keeps the number of states fixed, which JAX needs. The price is a small error in the momentum balance: mass leaves every node, not only the one at the drum. The error vanishes as $N \to \infty$, and the test suite checks that the release height converges: with 12 and 24 segments it differs by less than 2 %.

The resolution is set in **segments per km** of laid-out rope (`--segments-per-km`, default 10, so 100 m segments; at least 4 segments in total). Short and long ropes are then resolved alike. The release height is not very sensitive to the resolution. For the ASK 21 on the engine winch with 1200 m of Dyneema, 6, 12 and 24 segments give 463, 455 and 453 m. For the 10 km rope of Section 4.8, 20, 40 and 100 segments all give 2414–2415 m. A finer rope mainly shows better where the rope lifts off the ground and how it oscillates.

4.2 Segment tension

A rope pulls when stretched but cannot push. Its tension follows from how much each segment is stretched, plus a little internal damping that calms the oscillations along the rope.

For segment j, from node $j-1$ to node j, let $\boldsymbol{d} = \boldsymbol{p}_j - \boldsymbol{p}_{j-1}$ be the vector along it, $\ell = |\boldsymbol{d}|$ its length, $\hat{\boldsymbol{e}}_j = \boldsymbol{d}/\ell$ its direction and $\dot{\ell} = (\boldsymbol{u}_j - \boldsymbol{u}_{j-1}) \cdot \hat{\boldsymbol{e}}_j$ its rate of stretching. Then

$$T_j = \mathrm{softplus}_{2\,\mathrm{N}}\left(EA\frac{\ell - \ell_0}{\ell_0} + c\left(\dot{\ell} - \ell\frac{\dot{\ell}_0}{\ell_0}\right)\right), \quad c = 2\zeta\sqrt{EA\mu} \tag{11}$$

- EA: axial stiffness (Young's modulus times cross-section), in N;
- $\ell_0 = L_0/N$ and $\dot{\ell}_0 = -v_r/N$: rest length and its rate of change;
- $\zeta = 0.2$: damping ratio;
- the softplus (width 2 N) makes a slack rope carry almost no tension.

The damping term is proportional to the rate of *strain*, $\dot{\ell} - \ell\dot{\ell}_0/\ell_0 = \ell_0\, d(\ell/\ell_0)/dt$. So reeling in does not by itself create a damping force. With $c = 2\zeta\sqrt{EA\mu}$ each segment's stretching mode has damping ratio ζ, whatever its length. After release the top segment has a free end and carries no tension.

4.3 Aerodynamic drag (cross-flow principle)

A rope moving sideways through the air feels much more drag than one sliding along its own length. We use the cross-flow principle: split the air-relative velocity into a part across the rope and a part along it, and give each its own drag coefficient.

For segment j, the mean velocity relative to the wind is $\boldsymbol{v}_{\mathrm{rel}} = \frac{1}{2}(\boldsymbol{u}_{j-1} + \boldsymbol{u}_j) - \boldsymbol{w}$, with the wind at the segment's mid-height. Its part along the segment is $\boldsymbol{v}_t = (\boldsymbol{v}_{\mathrm{rel}} \cdot \hat{\boldsymbol{e}}_j)\hat{\boldsymbol{e}}_j$ and its part across is $\boldsymbol{v}_n = \boldsymbol{v}_{\mathrm{rel}} - \boldsymbol{v}_t$. The drag is

$$\boldsymbol{F}_{\mathrm{drag},j} = -\frac{1}{2}\rho d\ell\left(C_{Dn}|\boldsymbol{v}_n|\boldsymbol{v}_n + \pi C_f|\boldsymbol{v}_t|\boldsymbol{v}_t\right) \tag{12}$$

- ρ: air density at the segment's mid-height;
- d: rope diameter; ℓ: stretched segment length;
- C_{Dn}: cross-flow drag coefficient, about 1.2 for a cylinder;
- C_f: skin-friction coefficient along the rope, 0.01–0.02; the factor π turns the frontal area $d\ell$ into the surface area $\pi d\ell$.

Half of each segment's drag goes to each of its end nodes. Near the top of the launch the upper rope swings round the winch at high speed, so rope drag grows as the launch goes on.

4.4 Node equations

Each free node is pulled by the two segments on either side, feels half the drag of each, its weight and, while it lies on the runway, the ground:

$$m_n \dot{\boldsymbol{u}}_i = T_{i+1}\hat{\boldsymbol{e}}_{i+1} - T_i\hat{\boldsymbol{e}}_i + \frac{1}{2}\left(\boldsymbol{F}_{\mathrm{drag},i} + \boldsymbol{F}_{\mathrm{drag},i+1}\right) - m_n g\hat{\boldsymbol{z}} + \boldsymbol{F}_{\mathrm{gnd},i}, \quad \dot{\boldsymbol{p}}_i = \boldsymbol{u}_i \tag{13}$$

for $i = 1, ..., N-1$. Rope on the grass uses the same spring–damper and friction law as the glider's wheels (Equation 10). The spring is set by the node weight, $k_g = m_n g/s_g$ with $s_g = 2$ cm (the rope sinks 2 cm into the grass under its own weight), and the damping is critical, $c_g = 2m_n\sqrt{g/s_g}$. The friction coefficient μ_g is 0.5 for steel and 0.4 for Dyneema. Dragging 100 kg of steel rope over grass costs about 500 N during the ground run. Ground contact is checked at the nodes only.

4.5 Force on the glider

The rope acts on the glider through the hook. The force is the pull of the top segment, half its drag, the drag of the rope parachute and the weight of the rope end:

$$\boldsymbol{F}_{\mathrm{hook}} = -T_N\hat{\boldsymbol{e}}_N + \frac{1}{2}\boldsymbol{F}_{\mathrm{drag},N} - \frac{1}{2}\rho(C_DA)_{\mathrm{chute}}|\boldsymbol{v}_{a,h}|\boldsymbol{v}_{a,h} - m_{\mathrm{end}}g\hat{\boldsymbol{z}} \tag{14}$$

- T_N, $\hat{\boldsymbol{e}}_N$: tension and direction of the top segment (from node $N-1$ to the hook), so $-\hat{\boldsymbol{e}}_N$ points from the hook down along the rope;
- $(C_DA)_{\mathrm{chute}} = 0.1\,\mathrm{m}^2$: drag area of the closed parachute trailing in the rope end;
- $\boldsymbol{v}_{a,h}$: velocity of the hook relative to the air;
- $m_{\mathrm{end}} = m_{\mathrm{assy}} + \frac{1}{2}m_n$: end assembly ($m_{\mathrm{assy}} = 8$ kg: parachute, strop, weak link) plus half the top segment.

After release $\boldsymbol{F}_{\mathrm{hook}} = 0$. The **local** rope direction at the hook, from the hook to the last free node, is steeper than the straight line to the winch because the rope sags (Figure 1). Two angles are measured on this local direction: its angle below the fuselage axis, which trips the back-release when it exceeds 110°, and its angle below the horizontal, which triggers the optional pilot release (Section 7). The local direction also sets the hook moment (Equation 9). The pilot's attitude schedule and the winch driver do not use the rope angle; they use the elevation β of the glider's CG seen from the winch.

4.6 Validation: elastic catenary

A rope hanging still between two points takes the shape of an elastic catenary. Its exact solution [1] is a good check of the lumped rope. It is written in terms of the unstretched arc length s from the first end. The horizontal tension H is constant along the rope and the vertical tension is $V(s) = V_0 + ws$:

$$\begin{aligned} x(s) &= \frac{Hs}{EA} + \frac{H}{w}\left[\sinh^{-1}\frac{V}{H} - \sinh^{-1}\frac{V_0}{H}\right], \\ z(s) &= \frac{ws^2/2 + V_0 s}{EA} + \frac{H}{w}\left[\sqrt{1+\frac{V^2}{H^2}} - \sqrt{1+\frac{V_0^2}{H^2}}\right] \end{aligned} \tag{15}$$

where $w = \mu g$ is the rope weight per unstretched metre and V_0 the vertical tension at $s = 0$ (negative where the rope sags below its end). `cable.elastic_catenary` solves for V_0 and the unstretched length that fit the given end points. The test suite hangs a steel rope of 20 segments between two points 300 m apart and 80 m different in height, and checks that it settles onto the catenary to within 1 % of the sag.

A consequence visible in the launch results: in the climb, the tension at the hook is higher than the tension at the winch by roughly $\mu g \Delta z$, the weight of a rope as long as the height difference Δz (about 300 N for steel at 400 m).

4.7 Rope presets and rope input

Two generic presets are built in (Table 5). They are typical values, not a specific product.

Table 5: Generic rope presets (`cable.ROPES`, approximate values).

	steel wire	**Dyneema**
diameter d	4.5 mm	5.0 mm
mass per length μ	0.080 kg/m	0.016 kg/m
axial stiffness EA	1.0 MN	0.6 MN
breaking load	17 kN	27 kN
damping ratio ζ	0.2	0.2
cross-flow drag C_{Dn}	1.2	1.2
skin friction C_f	0.02	0.01
friction on grass μ_g	0.5	0.4
end assembly mass m_{assy}	8 kg	8 kg
parachute drag area $(C_D A)_{\text{chute}}$	0.1 m^2	0.1 m^2
sinking into the grass s_g	2 cm	2 cm

A specific rope can be given in three ways:

- **From datasheet values** with `cable.rope_from_datasheet`. It takes `diameter_mm`, `mass_kg_per_100m`, `breaking_load_kN` and either `elongation_at_break_pct` or `EA_N`, plus the optional `CDn`, `Cf`, `ground_mu`, `zeta`, `end_mass_kg` and `end_CdA_m2`. Without `EA_N`, the stiffness is the secant value $EA = F_{\text{break}}/\varepsilon_{\text{break}}$, which assumes a straight load–elongation curve. Synthetic ropes are stiffer at working loads, so give `EA_N` from the load–elongation curve at 20–30 % of the breaking load when you have it.
- **From a TOML file** (`--rope-file`), either with the same datasheet keys (example: `ropes/example_dyneema_6mm.toml`), or with `base = "steel"` (or `"dyneema"`) plus overrides of `Rope` fields in SI units (example: `ropes/example_steel_override.toml`).
- **Single fields on the command line**, in SI units: `--rope-param mu=0.09` (repeatable).

The breaking load does not enter the dynamics. It is only used to report the rope safety factor, F_{break} divided by the highest tension anywhere in the rope, in the summary table.

4.8 Long ropes: why the hook tension does not grow

Rope weight enters the tension balance through height, not length. Along a rope hanging in still air the tension changes by $dT/dz = \mu g$. The hook therefore feels at most the tension at the rope's lowest point plus $\mu g \Delta z$, which is small: $0.16\,\text{N/m} \times 3000\,\text{m} \approx 0.5$ kN for Dyneema. The rest of the rope's weight is carried by the winch end, by the sag, or by the ground. What a long rope does is **use up** the winch's pull before it reaches the hook: friction of the part still lying on the grass (up to $\mu_g m_{\text{rope}} g$) and drag of the long airborne part. Table 6 shows this for the ASK 21 on the engine winch.

Table 6: Long ropes: ASK 21, engine winch, 10 segments/km, no wind. Tensions are maxima while the glider is on the rope. [a]Share of the rope nodes on the ground at 60 % of the release height; for failed launches at the end of the ground run (120 s). [b]Lifts off later if the 120 s limit is raised (see text).

rope	**length**	**rope mass**	**release height**	**max T winch**	**max T hook**	**on ground**[a]
Dyneema	1.2 km	19 kg	455 m	6.2 kN	6.3 kN	0 %
Dyneema	10 km	160 kg	2414 m	5.6 kN	5.6 kN	2 %
Dyneema	50 km	800 kg	no lift-off[b]	5.6 kN	1.7 kN	99 %
steel	10 km	800 kg	no lift-off[b]	8.6 kN	3.0 kN	98 %
steel	50 km	4000 kg	no lift-off	7.0 kN	1.0 kN	100 %

A longer rope first gives more height: 10 km of Dyneema take the ASK 21 to 2414 m instead of 455 m. The hook tension does not grow with the rope; it is about the same as on the 1.2 km rope. But beyond

some length, the extra rope only adds friction and drag. At 50 km of Dyneema (800 kg) and at 10 km of steel (also 800 kg) almost all of the engine's pull is spent dragging the rope over the grass. Only a few hundred newtons reach the hook during the ground run. The glider rolls at about the reel-in speed, 70–85 km/h, well below the pilot's target speed. Whenever it hops off the ground, the pilot's speed correction lowers the nose again (Section 6), so it stays on the runway, and the launch counts as failed after 120 s (`no_liftoff`, Section 7). If this limit is raised, both do lift off in the end, after a very long ground run: 10 km of steel after about 175 s and 3.7 km, with release at 1540 m; 50 km of Dyneema after about 12 minutes and 17 km, with release at 2710 m. The 50 km steel rope (4 t, about 20 kN of friction on grass) cannot be dragged fast enough by this winch at all; the glider reaches only 28 km/h.

5 Winch

The winch turns engine power into rope pull. How the pull changes with reel-in speed decides how the launch feels: a constant pull, or one that fades as the rope speeds up.

Assumptions. The drum, gearbox and engine are lumped into one effective mass $M_{\text{eff}} = J/r_d^2$ at the rope (J the moment of inertia of the rotating parts, r_d the drum radius). The drum radius does not change as rope winds on. Both winch models share one equation for the reel-in speed:

$$M_{\text{eff}}\dot{v}_r = \tau(t,\beta)F_{\text{avail}}(v_r) - T_1 - bv_r, \quad F_{\text{avail}}(v) = \frac{P_{\max}}{\sqrt{v^2 + (P_{\max}/F_{\max})^2}} \tag{16}$$

- T_1: tension of the rope segment at the winch;
- $b = 20$ N s/m: viscous losses in the drum;
- F_{avail}: the pull the winch can give at full throttle. It is a smooth $\min(F_{\max}, P_{\max}/v)$: at low speed the pull is limited to $F_{\max}$ (torque limit), at high speed by the power $P_{\max}$;
- τ: the driver's throttle (Equation 17). It scales the whole pull curve, including the power limit.

The two winch types differ only in their parameters:

- **Tension-controlled winch** (`winch.tension_winch`): $P_{\max} = 10^{12}$ W, which is practically infinite, so the winch pulls with a prescribed force $F_{\max} = k_T mg$ (default $k_T = 1.1$, CLI `--pull`), whatever the reel speed. $M_{\text{eff}} = 60$ kg. This ideal winch isolates the glider's behaviour.
- **Engine winch** (`winch.engine_winch`): $P_{\max} = 200$ kW, $F_{\max} = 12$ kN and $M_{\text{eff}} = 250$ kg. The pull *drops* as the reel speeds up and *rises* when the glider slows the rope down (for example during the rotation into the climb). This is a real mechanism behind weak-link failures on light gliders. The driver sets the full throttle τ_0 so that the pull at 15 m/s reel speed is $k_T mg$ (a heavier glider gets more throttle), capped at $\tau_0 = 1$. For the ASK 21, $\tau_0 \approx 0.57$.

Winch driver. The driver opens the throttle smoothly over 3 s, starting at $t = 1$ s. Near the top the driver **throttles right down**: as the glider's elevation seen from the winch, β, goes from 65° to 70°, the throttle falls to `fade_floor` $= 0$ of full:

$$\tau(t,\beta) = \tau_0\, S\!\left(\frac{t-t_0}{t_r}\right)\left[1 - (1-f)\, S\!\left(\frac{\beta-\beta_0}{\beta_1-\beta_0}\right)\right], \quad \beta = \operatorname{atan2}(z - z_w, x_w - x) \tag{17}$$

with S the smoothstep, $t_0 = 1$ s, $t_r = 6$ s, $\beta_0 = 65°$, $\beta_1 = 70°$, $f = 0$ (`Winch.t_start`, `t_ramp`, `fade_beta0`, `fade_beta1`, `fade_floor`), and (x, z) the glider's CG. The ramp time follows tensiometer logs of real launches (`data/tensiometer`), in which the rope tension builds up over about 7 s. A much shorter ramp, or slack in the rope, lets the drum spin up before the glider moves. The drum then snatches the glider, and drum, rope and glider oscillate. With a 3 s ramp and 0.2 % slack, the hook tension of an ASK 13 on 1050 m of Dyneema (engine winch) swung between 1 and 6 kN in the ground run. The logs show a first peak of about 2.3 kN. At the top the drum stops pulling, the rope slackens

and its weight (plus the parachute drag) pulls the rope end down and back. The glider flies on over the rope end until the rope pulls from far behind and the hook's back-release trips (Section 7). After release the driver brakes the drum, $\dot{v}_r = -v_r/(2\,\mathrm{s})$, and the rope falls.

The driver's schedule is fixed in time and elevation. The driver does not react to the glider's speed or the rope tension. The placarded winch speed V_W is only reported, never enforced.

6 Pilot

What a real pilot does. During the ground run the pilot holds a little back stick, so the glider lifts off cleanly and does not nose over. After lift-off the glider must not climb steeply while it is still close to the ground, so the pilot lets it rotate into the climb gradually as it gains height. In the full climb the pilot watches the airspeed: when the glider is too fast the pilot pulls the nose higher, which puts more of the rope pull into climbing and slows the glider down; when it is too slow the pilot lowers the nose. Near the top, when the glider is high above the winch, the pilot lowers the nose. After release the pilot flies away in a normal glide.

What the model does. The pilot model has two parts. The first part says which pitch attitude the pilot wants, θ_{ref}. It is one smooth formula of the flight state, with no switches between phases. The second part moves the elevator to reach that attitude.

Assumptions (see also Section 1). The pilot is a simple, smooth control law, not a model of a human:

- The pilot knows the pitch attitude θ, the pitch rate q, the height h, the indicated airspeed V_{IAS} and the elevation angle β of the glider seen from the winch, exactly and without noise.
- There is no reaction time. The only delay is a first-order stick lag of 0.25 s.
- Every glider is flown with the same schedule. Only the climb attitude and the target speed differ between gliders (Table 7).
- The pilot does not use the release knob by default (see Section 7).

Target attitude. The target attitude is

$$\theta_{\mathrm{ref}} = \big(1 - s_{\mathrm{top}}\big)\big[\theta_g + s_{\mathrm{rot}}\big(\theta_{\mathrm{climb}} - \theta_g\big) + a(h)\,K_V\,c\big(V_{\mathrm{IAS}} - V_{\mathrm{target}}\big)\big] + s_{\mathrm{top}}\theta_{\mathrm{top}} \tag{18}$$

while the glider is on the rope, and $\theta_{\mathrm{ref}} = \theta_{\mathrm{glide}}$ after release. The symbols are:

- $\theta_g = \theta_{\mathrm{rest}} + 3°$: the ground-run attitude. This is the attitude at rest on the main wheel and tail, plus 3° of back stick.
- $s_{\mathrm{rot}} = S((h - 2\ \mathrm{m})/(30\ \mathrm{m} - 2\ \mathrm{m}))$: the rotation into the climb. It goes from 0 at 2 m height to 1 at 30 m. Here $S(x) = 3x^2 - 2x^3$ for x clipped to $[0, 1]$ is the "smoothstep" function, a smooth ramp from 0 to 1.
- θ_{climb}: the full-climb attitude, 35° or 40°.
- $c(\Delta V)$: the airspeed error $\Delta V = V_{\mathrm{IAS}} - V_{\mathrm{target}}$, limited softly to the range −4 to +2 m/s (the corners are rounded over 1 m/s). With $K_V = 2.5°$ per m/s the pilot therefore raises the nose by up to 5° when too fast and lowers it by up to 10° when too slow.
- $a(h) = \sigma((h - 0.5\ \mathrm{m})/0.1\ \mathrm{m})$, with σ the logistic sigmoid: a smooth "airborne" factor. It is 0 on the ground and 1 from about 1 m height, so the speed correction only acts in the air.
- $s_{\mathrm{top}} = S((\beta - 55°)/(70° - 55°))$: lowering the nose at the top. β is the elevation of the glider's centre of gravity seen from the winch (the straight line from the winch to the glider, not the rope direction). As β goes from 55° to 70° the target moves to $\theta_{\mathrm{top}} = 5°$.
- $\theta_{\mathrm{glide}} = 0°$: the attitude after release.

Elevator. The pilot moves the elevator like a PID controller (proportional, integral and derivative action), with a deflection limit and a lag:

$$\delta_{e,\mathrm{cmd}} = \delta_{\max} \tanh\left(\frac{-\left(K_p e + K_i e_I\right) + K_d q}{\delta_{\max}}\right), \quad \dot{\delta}_e = \frac{\delta_{e,\mathrm{cmd}} - \delta_e}{\tau_p}, \quad \dot{e}_I = a(h)e - (1 - a(h))e_I \tag{19}$$

Here $e = \theta_{\mathrm{ref}} - \theta$ is the attitude error and e_I its time integral. A negative elevator deflection (trailing edge up) pitches the nose up, hence the minus sign. The $K_d q$ term damps the pitch rate. The tanh limits the command smoothly to $\pm\delta_{\max}$, and τ_p is the stick lag. The integral only builds up in the air; on the ground ($a = 0$) it decays back to zero, so it cannot wind up while the tail is still on the grass. The default values are in Table 7.

Table 7: Pilot parameters (`gliders.make_pilot`). Per glider: Ka 8 35°, 85 km/h; ASK 13 35°, 100 km/h; LS4, Discus CS and ASK 21 40°, 105 km/h; ASG 29 and DG-1000 40°, 115 km/h.

symbol	default	meaning
$\theta_g - \theta_{\mathrm{rest}}$	3°	back stick in the ground run
h_{rot0}, h_{rot1}	2 m, 30 m	rotation into the climb starts, ends
θ_{climb}	35° or 40°	full-climb attitude (per glider)
V_{target}	85–115 km/h	target climb speed, IAS (per glider)
K_V	2.5° per m/s	attitude change per unit of speed error
β_{top0}, β_{top1}	55°, 70°	lowering the nose starts, ends
θ_{top}, θ_{glide}	5°, 0°	attitude at the top, after release
K_p, K_i, K_d	1.2, 0.8 s^{-1}, 0.6 s	elevator gains
τ_p, $\delta_{\max}$	0.25 s, 25°	stick lag, elevator limit
`release_angle`	180° (off)	optional pilot release (Section 7)
`back_release_angle`	110°	back-release of the hook (Section 7)

7 End of the launch: events

How a launch ends in reality. When the glider is high above the winch, the winch driver throttles back. The rope pull stops, and the pilot lowers the nose. The rope now hangs from the hook, and its weight and the drogue parachute pull the hook end down and back. When the rope pulls from far enough below and behind, the back-release in the tow hook opens by itself and drops the rope. The pilot can also pull the release knob at any time. If the rope pull gets too large, the weak link in the rope breaks.

How it ends in the model. The driver's throttle-down is part of the winch model (Section 5): as the elevation β goes from 65° to 70°, the throttle goes smoothly down to zero (idle). The rope slackens and sags, and the back-release trips. This is the normal end of a launch in the model, and it needs no decision by the pilot.

The launch ends at the first of the events in Table 8. Each event is a function of the time and the state that is negative during the launch; the event fires when it becomes positive (`no_liftoff` is a true/false condition instead).

If no event fires before the time limit $t_{\max}$ (Section 8), the event is reported as `t_max`. The *release height* in all results is the height of the glider's centre of gravity above its rest height, at the instant of the event.

What happens at the top. In all the default runs of Section 9 the launch ends with the back-release. Once the throttle-down starts at 65°, the glider gains only another 5–17 m. The back-release trips between 0 and 6 s after the glider passes 70° elevation, at a hook tension of 0.1–0.9 kN and 64–109 km/h IAS. There are two patterns:

- The Ka 8, ASK 13, ASK 21 and (on steel) DG-1000 still have time to lower the nose to 1–10° before the pull stops. The rope then has to sag to 100–109° below the horizontal before it is 110° below

Table 8: End-of-launch events, in the order of `simulate.EVENTS`. The first four (`LAUNCH_END_EVENTS`) end a launch normally.

event	condition
`release`	Pilot release: the local rope angle at the hook, φ_c, exceeds `Pilot.release_angle`. φ_c is measured *below the horizontal*, along the rope from the hook to the nearest rope node. Off by default (180°), because pilots cannot judge the rope angle; set e.g. 72–90° to model a pilot who releases on purpose near the top.
`back_release`	Back-release of the hook: the angle φ_b between the rope (hook to nearest node) and the fuselage axis, measured downwards, exceeds `Pilot.back_release_angle` = 110°. Roughly $\varphi_b = \varphi_c + \theta$. **The normal end of a launch.**
`weak_link`	The tension in the rope segment at the hook exceeds the glider's weak-link rating (the parachute drag and the rope-end weight are not included).
`rope_in`	Less than 30 m of (unstretched) rope is left out.
`no_liftoff`	Still less than 1 m above the rest position after 120 s, e.g. because the winch cannot drag a heavy rope. A failed launch: no sensitivities are computed.

the fuselage axis. This takes 1.5–2.5 s on Dyneema and about 5–6 s with the heavier steel rope. (With the engine winch the ASK 21 lies in between.)

- The LS4, ASG 29 and (on Dyneema) DG-1000 climb faster at the top, and their nose is still 24–30° up when the pull stops. With this attitude the rope is already 110° below the fuselage axis when it hangs 80–86° below the horizontal, so the back-release trips almost as soon as the throttle is closed, at about 70° elevation.

Optional pilot release. A pilot who releases at a set rope angle (`release_angle` < 180°) ends the launch slightly earlier and lower. For the ASK 21 on 1200 m Dyneema with the tension winch, a release at 72°, 76°, 80°, 85° or 90° gives 479, 481, 482, 484 or 486 m, against 489 m with the back-release. At 72° the glider is still being pulled (2.0 kN at the hook), because the driver has only just started to throttle down; from 80° on the hook tension at release is about 0.3 kN or less. The release height depends only a little on exactly when the launch ends near the top.

8 Solving it with diffrax

The idea. The whole system (glider, rope nodes, winch drum, pilot) is one set of ordinary differential equations, $\dot{y} = f(t, y, p)$, with the state y of Table 3 and the parameters p. It is solved with the JAX library diffrax. Three choices shape the numerics:

1. **Everything is smooth.** Every kink in the physics (rope going slack, ground contact, friction, stall, the pilot's schedule, the winch throttle) is rounded off with softplus, sigmoid, tanh or smoothstep functions. So there is a single continuous ODE for the whole launch, and an adaptive solver never has to stop at a discontinuity or switch between modes.
2. **Everything is a JAX function.** The vector field and all parameters are JAX pytrees. The whole solve can be compiled (`jit`), run for many launches at once (`vmap`), and differentiated (`grad`).
3. **The end of the launch is an event.** The solver finds the exact instant when an event function of Table 8 changes sign.

The details:

Vector field `dynamics.vector_field(t, y, args)`. `y` is an `equinox.Module` (`State`); `args` holds all parameters as `equinox.Module`s with float leaves (`params.py`) and a flag `attached` (1 on the rope, 0 after release).

Solver `Tsit5`, an explicit Runge–Kutta method of order 5. A `PIDController` sets the step size, with rtol = atol = 10^{-6}, `pcoeff` = 0.3 and `icoeff` = 0.4; the first step is 10^{-3} s. The fastest motion is the axial wave in the rope, with a frequency of about $\omega \approx \sqrt{EA/\mu}/\ell_0$, where $\ell_0 = L_0/N$ is

the rest length of one rope segment. For the Dyneema rope with 100 m segments this is about 60 rad/s at the start; it rises to about 150 rad/s near the top, as reel-in shortens the segments. The catalogue launches need 2200–4300 steps. The implicit solver `Kvaerno5` (`solver="kvaerno5"`) gives the same release height and time, but for the ASK 21 it takes 14 700 steps instead of 3100 and several times as long: the problem is not stiff enough for an implicit method to pay off.

Events `diffrax.Event` with the condition functions of Table 8, only for crossings from negative to positive (`direction=True`). An `optimistix.Newton` root finder (rtol = atol = 10^{-8}) locates the event time, and `sol.event_mask` says which event fired.

Two passes Pass 1 runs to the event and saves only the final state (`SaveAt(t1=True)`). So the time limit can be huge at no cost: $t_{\max} = 10^5$ s by default (`simulate.T_MAX`, `--t-max`), with at most $5 \cdot 10^6$ steps (`MAX_STEPS`). Pass 2 repeats the same, deterministic solve from 0 to the event time and saves 2000 evenly spaced points for plots and summaries. Long and short launches thus get the same number of points.

Free flight A second `diffeqsolve` starts from the state at release with `attached = 0` and runs for 8 s (200 points). The glider flies on with the glide attitude, the rope falls with a free end, and the winch driver brakes the drum.

Derived quantities Tensions, angles, load factor, power and so on are recomputed from the saved states with the same `evaluate` function that gives the derivatives, `vmap`-ed over time. So the plots can never disagree with the dynamics.

Gradients `sensitivity.release_height` is the same pass-1 solve (always with `Tsit5`) with `RecursiveCheckpointAdjoint` for reverse-mode differentiation. `eqx.filter_value_and_grad` over the whole `Launch` pytree gives the sensitivity to every parameter at once (Section 10).

Batching `simulate.solve_batch(launches)` stacks the parameter pytrees and `vmap`s the solve. The seven catalogue gliders run as one compiled program: about 5 s the first time (mostly compilation), then about 1 s.

9 Example results

Set-up. 1200 m rope, no wind, sea level with the standard atmosphere (ISA), no rope slack, rope modelled with 12 segments of 100 m (the default of 10 segments per km, `--segments-per-km`). The winch pull is 1.1 × the glider's weight. For the tension winch this is the constant pull $F_{\max}$. For the engine winch (200 kW, 12 kN) the driver sets the throttle so that the pull is 1.1 × weight at 15 m/s reel speed (Section 5). Reproduce Table 9 with

```
uv run main.py --no-sensitivity [--rope steel] [--winch engine]
```

Other options change the conditions: `--rope-length`, `--wind` (headwind at 10 m), `--pull`, `--field-elevation`, `--isa-dt`, `--segments-per-km`, `--t-max`, `--rope-file` and `--rope-param` (Section 4.7). Plots go to `results/`.

Observations.

- **Height.** On Dyneema with the tension winch the release heights are 457–494 m, 38–41 % of the rope length, after 42–54 s.
- **Steel** costs 2–24 m of height (most for the light Ka 8, least for the heavy DG-1000), gives a 5–11 m longer ground run, and lowers the rope safety factor: the steel rope is heavier and has a lower breaking load (17 kN against 27 kN). The DG-1000 on steel uses 42 % of the rope's strength.
- **Speed.** With the constant pull of the tension winch, the glider accelerates hard in the ground run and the early climb. The speed peaks about 11–12 s after the start, at 38–46 m height, during the rotation into the climb. This peak is above V_W for five of the seven gliders: by 28 % for the Ka 8, 23 % for the ASK 13, 15 % for the LS4, 9 % for the DG-1000 and 4 % for the ASG 29. The ASK 21 just reaches its V_W, and the Discus CS stays 3 % below it. A real driver would set a lower pull for the Ka 8 and the ASK 13.

Table 9: Launch results for the catalogue gliders, 1200 m rope, no wind. Every launch ends with the back-release. V_W: placarded maximum winch speed; h_{rel}: release height; time: launch time; roll: ground run to lift-off; max/min V: highest IAS, and lowest IAS once above 5 m; max T_{hook}: highest tension at the hook; max n: highest wing load factor L/W; max P: highest power at the drum; rope SF: breaking load divided by the highest rope tension. Bold: peak speed more than 1 % above V_W.

glider	V_W km/h	h_{rel} m	time s	roll m	max V km/h	min V km/h	max T_{hook} kN	max n	max P kW	rope SF
Dyneema, tension winch										
Ka 8	100	461	53.5	52	**128**	64	3.16	2.56	88	8.6
ASK 13	120	476	46.3	58	**148**	78	5.10	2.77	167	5.3
LS4	130	457	44.3	72	**149**	93	4.18	2.98	133	6.5
Discus CS	150	459	45.9	69	145	83	3.82	2.97	117	7.1
ASK 21	150	489	47.5	57	150	72	5.50	3.14	175	4.9
ASG 29	150	465	41.5	78	**156**	102	4.89	2.99	164	5.5
DG-1000	150	494	41.6	67	**164**	95	7.18	3.23	256	3.8
Steel, tension winch										
Ka 8	100	437	58.7	61	**120**	64	3.34	2.47	80	5.1
ASK 13	120	463	50.4	63	**143**	75	5.13	2.74	156	3.3
LS4	130	443	45.2	83	**144**	93	4.07	2.92	120	4.2
Discus CS	150	442	52.0	80	140	64	3.75	2.89	106	4.5
ASK 21	150	472	52.0	63	146	70	5.55	3.11	165	3.1
ASG 29	150	453	42.3	88	**153**	101	4.72	3.00	154	3.6
DG-1000	150	492	46.7	73	**161**	75	7.14	3.23	239	2.4
Dyneema, engine winch										
Ka 8	100	450	55.0	67	**106**	67	3.97	2.33	78	6.8
ASK 13	120	446	48.3	72	119	82	6.02	2.21	124	4.5
LS4	130	419	46.4	105	121	96	4.80	2.42	112	5.6
Discus CS	150	421	47.5	99	120	93	4.43	2.49	104	6.1
ASK 21	150	452	47.6	71	122	89	6.29	2.51	140	4.3
ASG 29	150	413	43.7	114	128	103	5.36	2.47	133	5.0
DG-1000	150	445	43.3	85	133	105	8.04	2.54	182	3.4

- **Engine winch.** The pull of the power-limited engine winch falls as the reel speeds up, so it limits the speed by itself. The peak speeds (106–133 km/h) and load factors (2.2–2.5) are lower, and only the Ka 8 is still above its V_W (by 6 %). The price is 11–52 m of release height; the fast gliders (ASG 29, DG-1000) lose most. The engine pulls harder when the glider slows the rope in the rotation, so the peak hook tension is 10–26 % *higher* than with the tension winch: about 80 % of the weak-link rating for the Ka 8 (4.0 of 5 kN), the LS4 (4.8 of 6 kN) and the DG-1000 (8.0 of 10 kN).
- **Load factor.** With the tension winch the highest wing load factor, 2.6–3.2, occurs at 31–34 m height, during the rotation, which the simple pilot flies rather briskly.
- **Top of the launch.** The lowest airspeed (64–102 km/h) is usually reached at the very end, as the glider slows after the throttle-down. For the Ka 8 this is 64 km/h, about 12 % above its 1 g stall speed in the model (57 km/h).
- **Power.** The ideal tension winch needs up to 256 kW at the drum for the DG-1000, more than the 200 kW of the engine winch.

10 Sensitivities of the release height

What is computed. How much would the release height h change if one parameter p changed a little? Every `uv run main.py` run (unless `--no-sensitivity`) answers this for all 81 model parameters at once: glider, rope, winch, pilot, environment and layout. It computes the derivative $\partial h/\partial p$ with one reverse-mode pass of `jax.grad` through the diffrax solve (`sensitivity.py`, `RecursiveCheckpointAdjoint`). The event time is found by the root finder, and diffrax differentiates it implicitly, so the change of

the release instant is included. The gradients agree with central finite differences to within 0.5 % (`tests/test_sensitivity.py`). Very small sensitivities are only as accurate as the solver tolerance, so this solve uses rtol = atol = 10^{-8}. The gradient takes about 10 s to compile and then about 1 s per launch; a whole `main.py` run for one glider takes about 30 s.

How to read the table. For each parameter the table gives its value, $\partial h/\partial p$ in metres per unit of the parameter (per degree for angles), the height change for a +10 % change of the parameter, and the *elasticity* $(p/h)\,\partial h/\partial p$: the percentage change of height for a 1 % change of the parameter. Rows are sorted by the size of the elasticity. All other parameters are held fixed. For example, the winch pull, preset as 1.1 × weight, does not follow a change of glider mass. These are local, linear estimates. `--only rope` restricts the printed table to one group, `--top` sets the number of rows, and the full table goes to `results/sensitivity_<glider>_<rope>_<winch>.csv`.

Table 10: The 12 largest sensitivities: ASK 21, 1200 m Dyneema, tension winch ($h = 489$ m). `uv run main.py --gliders "ASK 21" --top 12`.

parameter	name	value	$\partial h/\partial p$	Δh (+10 %)	elasticity
rope length	`rope_length`	1200 m	0.394 m/m	+47.2 m	0.97
winch pull	`winch.F_max`	5.07 kN	50.5 m/kN	+25.6 m	0.52
winch throttle	`winch.throttle`	1	256 m	+25.6 m	0.52
glider mass	`glider.mass`	470 kg	−0.48 m/kg	−22.5 m	−0.46
gravity	`env.g`	9.81 m/s²	−22.6 m/(m/s²)	−22.2 m	−0.45
end of nose lowering	`pilot.top_beta1`	70°	1.08 m/deg	+7.6 m	0.15
climb attitude	`pilot.theta_climb`	40°	1.51 m/deg	+6.0 m	0.12
target climb speed	`pilot.V_target`	105 km/h	−0.55 m/(km/h)	−5.8 m	−0.12
start of nose lowering	`pilot.top_beta0`	55°	0.81 m/deg	+4.5 m	0.09
back-release angle	`pilot.back_release_angle`	110°	0.38 m/deg	+4.2 m	0.09
rope diameter	`rope.diameter`	5 mm	−6.8 m/mm	−3.4 m	−0.07
rope normal drag coeff.	`rope.CDn`	1.2	−28 m	−3.3 m	−0.07

What the table shows.

- The rope length matters most. The height is almost proportional to it (elasticity 0.97): +120 m of rope gives +47 m of height.
- Next come the winch pull and the glider's weight, with about equal and opposite effect (+0.52 and −0.46): what counts is the pull per weight. `winch.F_max` and `winch.throttle` give the same row because the pull is their product, and `env.g` acts through the weight.
- Then the pilot. How late the pilot lowers the nose at the top (`top_beta0`, `top_beta1`), the climb attitude and the target speed each change the height by 4–8 m for a 10 % change. So does the angle at which the back-release trips. These control-law parameters matter more than any single aerodynamic parameter of the glider: the largest, the wing area, comes next with +3.1 m for +10 %.
- Rope drag (diameter and normal drag coefficient) costs about 3.3–3.4 m each for +10 %.
- The driver's throttle-down angles have small effects: moving the start from 65° to 71.5° (+10 %) costs 2.1 m.

Parameters whose default value is zero have zero elasticity and so do not appear at the top, although their derivative can be large. The most important one is the headwind: $\partial h/\partial W_{\text{ref}} = 22$ m per m/s of headwind at 10 m height. Another is the throttle floor after the throttle-down (`fade_floor` = 0): keeping 10 % of the power would add about 5 m (linear estimate).

The rope. For the steel rope with the ASK 13 (`--rope steel --only rope`, $h = 463$ m) the rope properties change the height by, for +10 %: diameter −2.4 m, normal drag coefficient `CDn` −2.3 m, mass per length −1.4 m, parachute drag area −0.5 m, ground friction −0.4 m, rope-end mass −0.4 m, tangential friction `Cf` −0.1 m. Rope drag and rope weight thus matter about equally, with drag somewhat ahead. For the much lighter Dyneema rope the weight hardly matters (−0.2 m for +10 % on

the ASK 21). The axial stiffness EA has essentially *no* influence on the release height: +10 % changes it by about 1 cm. Rope stretch shapes the tension oscillations in the ground run (Section 4), not the energy that reaches the glider.

11 Limitations and possible extensions

- **Pilot and driver.** Both are simple, smooth control laws with perfect information and no reaction time (Section 6, Section 5). The driver works open loop: the throttle depends only on time and on the glider's elevation, not on its speed or the rope tension. The driver does not react to overspeed; V_W is only reported, not respected. The launch ends by the back-release after the throttle-down on a visual cue; a pilot release when the pull fades (a tension cue) would be an alternative. As Section 10 shows, the parameters of these control laws affect the release height more than any single aerodynamic parameter, just as pilot and driver technique do in real life.
- **Gliders.** The glider data are approximate estimates, not manufacturer data. The model is longitudinal only: no wing drop, roll or yaw. There is no ground effect and no flaps (flap settings matter for the ASG 29 and similar gliders).
- **Rope end.** During the launch the parachute is a fixed, closed drag area plus a mass at the hook. After release both disappear: the rope falls with a bare free end. The parachute does not open.
- **Reel-in.** Reeling in removes mass evenly from all rope nodes (Section 4.1). A "moving-node" formulation that removes mass only at the drum would be exact.
- **Winch.** The drum physics is intentionally minimal (one effective mass).
- **Atmosphere.** Steady wind with a power-law profile only; no gusts or thermals.
- **Sensitivities** (Section 10) are local. For large changes (e.g. steel to Dyneema) run the simulation itself. The same gradients could also drive an optimisation of the tension profile or the pilot schedule, subject to $V \le V_W$, $T \le$ weak link and $n \le n_{\max}$ [2].

12 Related literature

Winch-launch simulation. Gäb and Santel [3] (RWTH Aachen, Matlab/Simulink) model the aircraft, pilot, winch, winch operator, cable, atmosphere and terrain. Their cable is a chain of point masses joined by spring–damper links, with drag, weight and ground reaction: the same approach as Section 4 here. Their pilot and winch operator are PID controllers with reaction time and neuromuscular delay. They study a reference launch, wind, and too steep initial climbs. **It is the best reference to validate this model against.** It grew out of an RWTH student thesis [4] (CORE PDF). That thesis found that opening the throttle quickly destabilises the phugoid (the slow pitch–speed oscillation), that the tow-hook position can be optimised, and that the release height depends on cable type, tow distance, maximum winch force and especially cable drag. It lists ground effect and ground/cable friction as missing. Santel's diploma thesis [5] studies winch-launch accidents with multipoint aerodynamics. Bogan's simpler model [6] (prescribed flight path, MathCAD, not peer-reviewed) gives a good feel for how the cable drag grows as the rope turns broadside to the flow.

Optimal launch trajectories. With limits on the lift coefficient and on the cable acceleration, the optimal trajectory is climb–dive–climb; a limit on the reel-in speed makes it more realistic [2]. This is the benchmark for an optimisation with `jax.grad` (Section 11). See also Eppler on the optimal release height [7] and the follow-up work of Pierson and Chen [8].

Safety and winch engineering. Browning [9] and Hills [10] on safe winch launching; design formulae for winch hardware [11]; the Soaring Safety Foundation notes by Smith [12] (Winches).

Cable and tether modelling. Irvine's elastic catenary [1] is used to validate the rope model in Section 4.6. Modelling of tethers with variable length [13] relates to the reel-in approach of Section 4.1; the SkySails model [14] does optimal control with tether force and reel-out speed as variables.

How this model compares with [3]. The structure is the same: a lumped-mass cable, a PID pilot and a tension-controlled winch operator. The differences: [3] includes human reaction and neuromuscular delays and a more detailed winch and operator model. This model keeps the number of rope segments fixed during reel-in, uses smooth contact models (one continuous ODE instead of switching between modes), and uses JAX/diffrax for batching and gradients. Obvious next steps: reproduce the reference launch of [3], and add reaction delays to the pilot and the winch driver.

Bibliography

[1] H. M. Irvine, *Cable Structures*. MIT Press, 1981.

[2] "Maximum Altitude Sailplane Winch-Launch Trajectories," *Aeronautical Quarterly*, vol. 28, no. 2, pp. 75–84, 1977, doi: 10.1017/S0001925900007976.

[3] A. Gäb and C. Santel, "Numerical Simulation of Glider Winch Launches," *Technical Soaring*, vol. 35, no. 3, 2011, [Online]. Available: https://ts.ostiv.org/index.php/ts/article/view/76

[4] "Numeric Simulation of a Glider Winch Launch," 2008. [Online]. Available: http://publications.rwth-aachen.de/record/230129/files/3265.pdf

[5] C. Santel, "An investigation of glider winch launch accidents utilizing multipoint aerodynamics models in flight simulation."

[6] L. Bogan, "Glider Winch Launch Simulation." [Online]. Available: https://bogan.ca/soaring/winch/winch.html

[7] R. Eppler, "Windenschlepp und optimale Ausklinkhöhe."

[8] Pierson and Chen, "Follow-up work on optimal sailplane trajectories."

[9] H. Browning, "Boundaries of Safe Winch Launching," *Technical Soaring*, vol. 31, no. 4, p. 95, 2007, [Online]. Available: https://soaringweb.org/Soaring_Index/Technical_Soaring/Technical_Soaring_issue.html

[10] T. Hills, "Safety Analysis of the Winch Launch," *Technical Soaring*, vol. 31, no. 4, p. 101, 2007.

[11] "The Design and Development of Glider Launching Winches," *Technical Soaring*, [Online]. Available: https://ts.ostiv.org/index.php/ts/article/view/942/0

[12] B. S. Smith, "Winches; Winch launching revisited." [Online]. Available: https://www.soaringsafety.org/publications/Winch-launching-revisited.pdf

[13] Williams, Lansdorp, and Ockels, "Modeling and control of a kite on a variable length flexible inelastic tether," AIAA-2007–6705, 2007.

[14] "A quaternion-based model for optimal control of the SkySails airborne wind energy system." [Online]. Available: https://arxiv.org/abs/1508.05494